

Business Process Redesign of Sales Monitoring Business Process Management to be a Smart Logistic Shipment Company

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The sales process in every company is important. Therefore, every sales record must be stored in an integrated database so that tracking sales records is easy. However, there are still many sales recording practices that are still done manually, making the process of recording sales complicated and time-consuming. Therefore, it needs a sales record automation process that can track all sales recording processes that provide standardization in the form of KPIs and progress at each stage of sales records. With the growth of smart logistics, it can drastically change the business process of dashboard monitoring in a logistics company. With the smart logistic concepts and the use of business process redesign methodology, it resulted in a more simplified yet integrated automated sales monitoring system.

Keywords— Sales automation, standardization, KPI, tracing, sales business process

I. INTRODUCTION

Business Process Redesign (BPR) is a series of businesses that are rethought as well redesigned to achieve an improvement in the business chain. BPR aims to figure out how to create an entire set that exists in an organization or company increased. Realizing the potential benefits to be gained from implementing BPR, many logistic companies in Indonesia are trying to improve themselves to become more competent companies by implementing BPR [1].

BPR implementation in a company is worth considering even though, based on data, many organizations have failed to implement BPR. Implementing BPR allows companies to improve themselves to increase competitiveness against competitors [2]. If it runs successfully, BPR can improve the overall objective to become more effective and efficient in an organization's domain such as quality of services, products, and minimize cost. Because of that BPR is a tool of choice for maintaining a continuation of competitiveness in advantage and ensuring the efficiency either in the private or public sectors [3]. To increase the chance of BPR running successfully, it must have collaborated with good business process management (BPM).

BPM is a technique for coordinating business activities to increase organizational effectiveness [4]. BPM is a set of

ideas, instruments, and methods that improve business process design, administration, configuration, and analysis. Sales force automation is one of the technologies to assist BPR adoption in Sales Force Monitoring (SFA). Due to the fact that SFA is a complementary digital sales tool that is increasingly important for business-to-business organizations looking for organizational efficiencies and the production of customer value [6]. In a logistics company, it is important to digitalize the shipping and logistics process. In order to compete in this marketplace sector, multinational companies must determine their core competencies and highlight their value-added logistics activities to third-party logistics providers in order to have higher advantages to compete in this market [7]. With the rapid growth of smart logistics era where it increases the effectiveness and capacity of the logistics service, many logistic services are in a similar scenario where their monitoring logistic supply chain is not optimized with their fast-moving product which affect their monitoring system is not in real time [8].

There are several success factors in BPR that have been grouped into five categories, including cross-functionality from each team, the implementation process used by the project team, the team's access to knowledge regarding the processes being redesigned or reengineered, the caliber of the IT support provided for the project, and project leadership and motivation [9]. Because of that it is noted that a significant amount of work entails the integration of business processes and information technology. To enable the speeding of current business operations, improvements were implemented by using the information systems that focused on sales monitoring by using Salesforce to process and organize sequentially in accordance with each method's tasks [10].

II. LITERATURE REVIEW

Many authors agree by considering BPM as an important method to be implemented to improve business performance, competitiveness, and flexibility. By using BPM, a company can assure that its processes deliver on their promise and operate [11]. It also can bring advantages, such as improvement of business processes, quality of performance information, and a clearer vision of the organization about

their roles and goals needed to achieve [12]. By using BPM, firms can use new technology integration to speed up analysis and enhance the existing business process in numerous ways. For instance, by using process mining technology for the procurement process where the owner of the process can immediately identify the as-is process consistently to the extent that they can do so, as well as process issues such as the quantity and value of rework activities such as price changes [13].

To help the business process redesign, it needs the concepts of smart logistics. The collaboration of logistics and the supply chain technology is becoming more important due to the swift development of the commodity economy. To assess this situation, the logistics industry applied intelligent information technology, such as blockchain and big data analysis, the using of automation for visualization, tracing, and intelligent decision-making of the logistics business process [14]. Several countries have adopted a smart logistic policy to standardize the logistic industry and become more traceable, below is the list of the countries.

TABLE I. SMART LOGISTIC COUNTRY POLICY [8]

Country	Policy	Focus
United States	Commodity Flow Survey (CFS)	For the purposes of transportation planning and determining the demand for transportation facilities, data on commodity movement are collected.
United Kingdom	Industrial strategy: Artificial intelligence sector deal	Development of AI and data-driven economics
France	Logistics Strategic Plan 2025	Development of innovation in logistics and the collaboration of smart logistic and smart manufacturing
European Union	The Trans-European Transport Network (TEN-T) policy	Development of a wide transportation network by using state-of-the-art technologies
Russia	The State of the Union Address 2018	Create a digital platform that is compatible with the world's information space.
Japan	Strategic Innovation Promotion Program (SIP) phase two	Develop a self-driving system
China	Opinions on promoting high-quality development of the logistics industry and facilitating the formation of a strong domestic market	Create a high-quality network of logistics facilities and a shared logistics information platform to help the logistics industry aid the real economy with services.

In developed country list as above, they already implemented a policy as standardization for all logistic companies, all those policies have integrated the information flow as an end-to-end process. It visualizes below.



Fig. 1. Smart Logistic Concepts [8]

As shown in Fig. 1, the differences between the traditional logistic company, is the using of technologies such as the Internet of Things (IoT) and big data analytics to ensure the information flow is integrated. First, advanced technologies used in intelligent systems, include artificial intelligence (AI), automation, and information and communications technology (ICT). These technologies are used throughout the logistics process to raise the level of automation for logistical operations and enable wise decision-making for typical logistics management issues.

The second is flexibility. Due to better inventory optimization, more precise demand forecasts, and more effective transportation planning, smart logistics provides greater flexibility. Smart logistics' growing capacity to address unforeseen difficulties improves customer satisfaction.

The third is the integration of logistics. Information sharing amongst logistics process agents is made possible by technologies like IoT and ICT, and related business processes can be centrally handled, enhancing the coordination of various logistical operations.

Fourth is Self-organization. The logistics system can operate with little to no human involvement thanks to real-time monitoring and intelligent decision-making, which increases the efficiency of logistical operations. Because many developed countries have implemented a comprehensive smart logistic policy, a developing country like Indonesia should follow the creation of the same policy by taking a small step to create a big policy. The first step that can lead Indonesia to have a standard policy is to create a monitoring automation system for sales business process. By using a digital sales tool to automatically be monitored, it can change the dynamics and work division, and power relations, between the selling and buying organizations [15]. For instance, most of the simple sales tasks, such as "explaining" and "order taking", become obsolete, as e-commerce takes over order taking, and marketing automation takes over explaining [16].

III. METHODOLOGY

This study employs a unique technique for BPR-based business process redesign. BPR can assist researchers in achieving their goals, which may include raising efficacy and efficiency and lowering overhead costs. BPR ought to employs information technology to support new business processes, attempts to accomplish organizational-level strategic results, and is frequently inter-functional in its efforts. Involves a radical restructuring of business processes [17]. The BPR methodology adopted framework for this research is visualize as below.

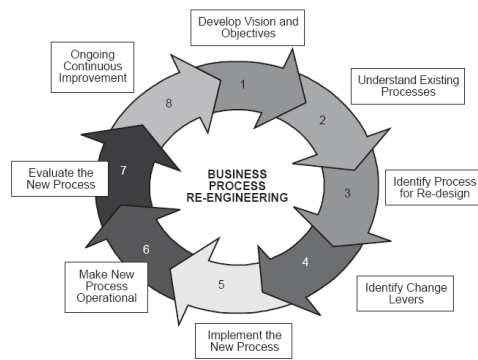


Fig. 2. BPR Methodology [17]

As shown in Fig. 2, this methodology is a life cycle methodology where it can be implemented over and over to enhance the business process. The first step is to develop vision and objective. Researcher must have a clear vision regarding about the sales monitoring and its objectives to create and automated sales monitoring process. Second, is understanding the existing processes. To create an as-is solution, researcher must understand the problem happened in the existing business process to create a more simple and effective solution. Third, is identify the process for redesign. In this step researcher create a mapping which process should be removed, merged, or added to create a new business process for sales monitoring. Fourth, is identify the change levers. In this step the problem must be listed as issues to help the creation of the new process. Fifth, is to implement the new process. In this step the redesigned business process is implemented in a sandbox to create a simulation how the end-to-end process is done. Sixth, the process must be put into operational to evaluate the new business process before it enters the production. Seventh, the process will be evaluated by using black box testing to test the functionality of the software itself by the user. The last step is ongoing continuous improvement to add more values in the business process itself if there is a new founding.

IV. RESULT AND DISCUSSION

By develop vision and objectives, it must align with the output of the company. The output in this business process reengineering method is the sales monitoring process can be automatically recorded. After that, the existing problem must be created in a understand manner to mapping the problem in the existing process, below is the existing business process.

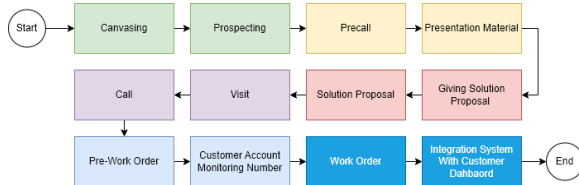


Fig. 3. Existing Business Process

This existing business process still done manually because is use many databases that still not integrated. The color different in the existing business process indicating that that process is saved in different database, so it only integrated in the end process to avoid any integration mismatch. This existing business process resulting a major issue in the sales monitoring because there is no standardization in customer view and no real time record from the sales division because the integration is not in real

time. Below is the list of issue and user that affected by using expert judgment process. By using expert judgement process, it can list the prioritize problems by the expert user [18].

TABLE II. LISTED ISSUES

<i>Problem</i>	<i>Affected User</i>
Record process still done manually	Sales division
No customer profile for the easiness of follow up by sales team	Sales division
Daily activity record is hard to trace because the recor process is done manually	Sales division
The integration is happened in the end of the process and still done manually by using exported excel document	Sales division
The KPI target is done manually and inconsistent	Sales division
Customer have variety dashbord which makes harder to understand from one sales to another sales	Sales division and customer

From the listed issues above, a redesign business process is created and implementing a sales force automation tool. When implementing sales force automation tool, there are several changes in the existing business process, it visualizes as below.

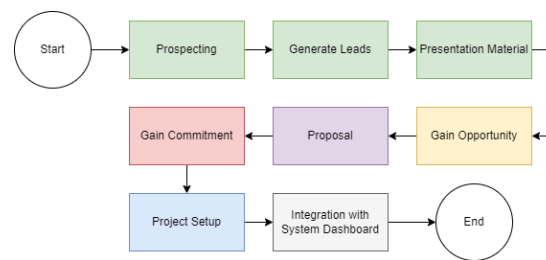


Fig. 4. Targeting Business Process

The result of identification of the problem, resulting the targeting business process that have been aligned with sales force automation tools. With this solution, it is resulting a more beneficial advantages that will be listed below.

After the redesign business process is created, it become operational and tested in a sandbox before putting into production. The flow process of the redesigned business process is visualized as below.

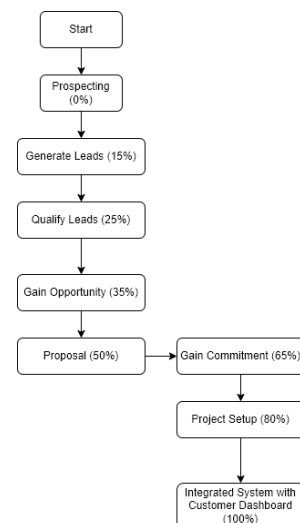


Fig. 5. Redesigned Business Process Flowchart

Based on Fig. 5 the redesigned business process has a KPI and SLA for each process and can be traced by the sales division from 0% until the process is finished which is 100%.

Black box testing will be utilized as the evaluation method to test the functionality of the created monitoring program. hence this evaluation method concentrates on what the software does [19]. There are stages of input, procedure, and output in black box testing. The process column and system task in the table will provide an explanation of the processes that are performed. The predicted output values are the success indicators, while the input values are entered in the testing procedure column. The black box testing table, which was based on user testing, is shown below.

TABLE III. BLACK BOX TESTING

<i>Process</i>	<i>System Task</i>	<i>Testing Procedure</i>	<i>Expected Output</i>	<i>Status</i>
Account Creation	Create an account by going to the account tab to create a new account	If the account is already created it shows an alert	It can create an account	Success
Contact Creation	Create a contact in an account that has previously been created	If the contact is created, it can generate a lead	The account that has contact is converted to a lead	Success
Leads Creation	Lead conversion is carried out if the lead is qualified	If the lead is not qualified, you must enter the reason for the lead	If the lead is successfully converted, it will become an opportunity. If the lead is not qualified it will be a recorded in database as a future lead	Success
Opportunity Creation	Create new opportunities based on qualified leads	The leads must be automatically in opportunity creation list	The opportunity can be created if the user input all the required fields	Success
	Validation loss for unfavorable opportunity	The opportunity can be closed before creating a proposal	The opportunity is changed become lost opportunity	Success
	Gaining Commitment to approved a quotation	Sales will send quotes to be approved based on the discount given	The quotes will be sent based on the discount given, if the discount is 15% then sales lead and head of	Success

<i>Process</i>	<i>System Task</i>	<i>Testing Procedure</i>	<i>Expected Output</i>	<i>Status</i>
			branch must approve the quotation	
Proposal Creation	Make a proposal based on the quote that has been approved	The proposal can be completed until it chooses the closed stage	The proposal must be syncing with opportunity to monitor the sales until it closes	Success

With this redesigned business process each activity is recorded automatically by using sales force automation tools. For the detail listed benefits based on the expert judgment process is listed as below.

TABLE IV. LISTED BENEFITS

<i>Benefits</i>	<i>Affected User</i>
Most problem is answered by the paint point sales of customers	Sales division and customer
Sales tool standarization	Sales division
Clear regulation and work order	Sales division
KPI is defined and standardized	Sales division
Daily activity record is recorded and can be monitored in one system	Sales division
All sales activity is recorded in one system, so the integration is done automatically by sales force	Sales division and customer
Forecast target revenue become available	Sales division

This benefit is resulted using sales force automation tools. This study offers important managerial insights for the creation of sales processes and associated tools, as well as for redefining the role of a salesman, which is in line with the findings of other researchers. First, as purchasing behavior shifts toward more autonomous collecting of service and product information, digitalization fundamentally alters purchasing processes which is aligned with other researchers [20] [21]. Second, the use of sales configurators has significant ramifications for the conventional approaches of carrying out a salesperson's duties. For instance, by using sales configurators to complement this process, the pre-call and presentation material processes can be combined into one. This results are in line with the BPR success factor which is consist of the cross-functionality of the project team and quality of the IT support extended to the project.

For the continuation of improvement this sales automation process, there are several aspects to create a more efficient of sales automation, it is shown below.

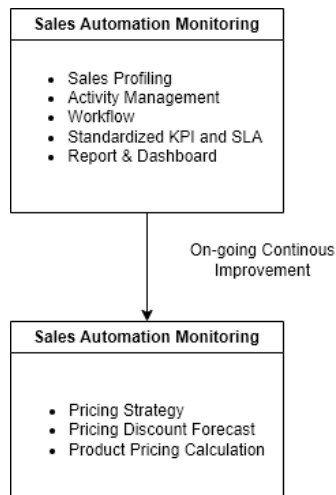


Fig. 6. On-going Continuous Improvement

In Fig. 6 the on-going process to improve the sales automation is pricing strategy, discount forecast, and pricing calculation. The reason why those three components is on-going improvement because it is needed to calculate the process in prospecting between the logistic company and the customer to achieve the desired pricing.

V. CONCLUSION

In this research, the result of the business redesign process is considered successful by the supporting evidence of other hypotheses which perceived enjoyment affecting perceived usefulness. The result of this redesign process is aligned with the smart logistic concept which is using an integrated big data for information flow start to end.

Although the implementation is only for sales monitoring, it is one step closer to create a concept for smart logistic in Indonesia. The adoption of new tools or systems and perceived usefulness are further questions raised by this conclusion. Like all studies, this research has limitations that present intriguing directions for future investigation. This research only focused on the sales activity monitoring to reduce the manual record process and manual integration and the sample used in the study came from a single organization.

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